An Adaptive EEG-to-Spike Framework for Energy-Efficient Neuromorphic Motor Imagery Decoding: A Survey and Proposed Methodology

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Abstract

Motor imagery (MI) based Brain-Computer Interfaces (BCIs) have emerged as a transformative technology for enabling direct communication between the human brain and external devices, with critical applications in assistive technologies for paralysed patients and neural rehabilitation. Conventional MI-BCI systems rely on deep learning architectures such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, which are computationally expensive, energy-hungry, and therefore incompatible with wearable or implantable neuromorphic hardware. This paper surveys the current state of EEG-based motor imagery decoding with a focus on five critical and underaddressed research gaps: (1) energy inefficiency of conventional deep learning BCIs, (2) subject-agnostic fixed frequency band selection, (3) manually engineered spike encoding thresholds, (4) single temporal scale processing in spiking neural networks, and (5) absence of standardised energy evaluation metrics. To address these gaps, we propose an Adaptive EEG-to-Spike Framework comprising three novel contributions: per-subject adaptive frequency band selection using KNN-based Mutual Information (KNN-MI) estimation, end-to-end learnable spike encoding via surrogate gradient descent jointly optimised with a spike-count sparsity penalty, and a multi-timescale Parametric Leaky Integrate-and-Fire (PLIF) spiking neural network with parallel fast and slow temporal branches fused by an attention mechanism. Evaluated on the PhysioNet EEG Motor Movement/Imagery dataset (103 subjects, 64 channels, 160 Hz), the proposed framework targets 4-class MI classification with synaptic operation (SynOps) counting as a standardised energy efficiency proxy. Comparative benchmarks from peer-reviewed literature (2020–2025) support each proposed component, with related adaptive selection methods demonstrating up to 21.76% accuracy improvement over fixed-band baselines and SNN deployments on neuromorphic hardware achieving up to 95% energy reduction compared to equivalent DNN implementations.

Index Terms — Brain-Computer Interface, Motor Imagery, EEG Decoding, Spiking Neural Networks, Parametric LIF, Surrogate Gradient, Adaptive Frequency Selection, Neuromorphic Computing, Energy Efficiency, Synaptic Operations.

1. Introduction

Brain-Computer Interfaces (BCIs) establish a direct communication pathway between the human brain and external computational devices without requiring any peripheral muscular activity [1]. Among the various BCI paradigms, Motor Imagery (MI) has attracted significant research attention due to its non-invasive nature, relative ease of signal acquisition, and clinical relevance for patients suffering from paralysis, stroke, or amyotrophic lateral sclerosis (ALS) [2].

In an MI-BCI system, the subject imagines performing a physical movement — such as opening the left hand or lifting both feet — without executing any actual motion. The resulting modulations in the electroencephalogram (EEG) signal, particularly in the μ (8–12 Hz) and β (13–30 Hz) frequency bands, manifest as Event-Related Desynchronisation (ERD) at movement onset and Event-Related Synchronisation (ERS) at movement offset [2]. These modulations serve as the discriminative features for classifying the intended movement.

Contemporary MI-BCI pipelines predominantly employ deep learning architectures including Convolutional Neural Networks (CNNs) [3] and LSTM networks for feature extraction and classification. While these methods achieve competitive accuracy, they impose substantial computational burdens, consuming hundreds of millijoules per inference [4]. This renders them fundamentally incompatible with wearable, edge-deployable, or implantable BCI hardware, which typically operates under strict power budgets in the microwatt range.

Spiking Neural Networks (SNNs), inspired by the information processing principles of the biological brain, offer a compelling alternative. Through event-driven, sparse spike-based computation, SNNs inherently reduce the number of active multiply-accumulate (MAC) operations, enabling up to 95% energy reduction com-

pared to conventional DNNs when deployed on dedicated neuromorphic hardware such as Intel's Loihi [5].

Despite this promise, existing SNN-based BCI systems still suffer from several critical limitations. They apply identical fixed frequency bands to all subjects, ignoring well-documented inter-individual variability in EEG spectral characteristics [6]. Spike encoding thresholds are manually engineered rather than learned from data [5]. All existing SNN-BCIs use a single membrane time constant, failing to capture the dual timescale dynamics inherent in MI-EEG signals [11]. Finally, energy efficiency claims are rarely supported by standardised, reproducible metrics [21].

This paper makes the following contributions:

- A systematic survey of the current EEG-BCI literature identifying five critical research gaps with quantitative benchmark evidence.
- A proposed adaptive EEG-to-spike framework that addresses all five gaps through three novel architectural contributions.
- A comprehensive comparison of the proposed framework against established baselines on the PhysioNet Motor Imagery dataset.

The remainder of this paper is structured as follows. Section 2 provides background on EEG signal characteristics and SNN fundamentals. Section 3 surveys related work. Section 4 identifies and quantifies the five research gaps. Section 5 describes the proposed methodology. Section 6 outlines the evaluation protocol. Section 7 discusses implications and limitations. Section 8 concludes.

2. Background

2.1 EEG-Based Motor Imagery

Electroencephalography (EEG) measures the aggregate electrical activity of cortical neurons via electrodes placed on the scalp surface. The resulting signals are characterised by low amplitude (5–100 μV), susceptibility to artefacts from ocular and muscular sources, and non-stationarity across both time and subjects [23].

For motor imagery tasks, the primary discriminative information resides in two oscillatory rhythms generated in the sensorimotor cortex. The μ rhythm (8–12 Hz) originates from the sensorimotor cortex with bilateral foci of desynchronisation over the hand area. The β rhythm (13–30 Hz) exhibits a distinct topography with peak desynchronisation over the vertex and is associated with sustained motor states [22]. Critically, these two rhythms are neurophysiologically independent processes [22], arising from separate cortical generators and exhibiting different temporal dynamics, with ERD in the μ-band beginning approximately 2 seconds before movement onset, while β-band ERS develops gradually 500–1500 ms after movement termination.

2.2 Frequency Band Variability Across Subjects

A well-established limitation of fixed-band approaches is the substantial inter-individual variability in EEG spectral characteristics. Peak frequencies in the μ and β bands vary by 2–5 Hz across subjects due to differences in cortical anatomy, skull thickness, and electrode placement [6]. Consequently, a fixed band centred at 10 Hz may capture the peak activity of one subject while completely missing the dominant frequency of another.

2.3 Spiking Neural Networks

SNNs constitute the third generation of artificial neural networks [29], modelling the temporal spike-based communication of biological neurons. The Leaky Integrate-and-Fire (LIF) neuron, the most commonly employed SNN model, integrates incoming weighted spike inputs as a change in membrane potential v_j^t according to:

$$v_j^{t+1} = \alpha v_j^t + \sum_i W_{ji} z_i^t + \sum_i W_{ji}^{in} x_i^t - z_j^t v_{th} \qquad (1)$$

where $\alpha = e^{-\delta t/\tau_m}$ is the membrane decay factor, v_{th} is the firing threshold, z_i^t are presynaptic spike inputs, and x_i^t are external inputs. The neuron emits a spike $z_j^t = H(v_j^t - v_{th})$ when the membrane potential exceeds the threshold, where $H(\cdot)$ is the Heaviside step function.

The energy efficiency of SNNs stems from the sparse nature of spike-based computation. In contrast to conventional ANNs, where every neuron produces an output at every timestep, SNN neurons are silent for the majority of computation time, firing only upon threshold crossing. This sparsity directly reduces the number of synaptic multiply-accumulate (MAC) operations, quantified as Synaptic Operations (SynOps) [24]:

$$\text{SynOps} = \sum_\ell n_\ell^{out} \cdot \bar{s}_\ell \qquad (2)$$

where n_ℓ^{out} is the fan-out (number of output connections) of layer ℓ and $\bar{s}_\ell$ is the average number of spikes emitted per neuron in that layer.

2.4 Training SNNs with Surrogate Gradients

A fundamental challenge in SNN training via backpropagation is the non-differentiability of the Heaviside step function $H(\cdot)$, which has a zero derivative almost everywhere. SLAYER [13] introduced the surrogate gradient approach, replacing the true derivative with a smooth approximation during the backward pass. SpikingJelly [12] implements the arctangent (ATan) surrogate:

$$\frac{\partial H}{\partial v} \approx \frac{1}{\pi} \cdot \frac{1}{1 + (\pi v)^2} \qquad (3)$$

This approximation has a peak at $v = 0$ and decays smoothly away from the threshold crossing point, enabling gradient-based optimisation of deep SNN architectures.

3. Related Work

3.1 Deep Learning for EEG-BCI

The seminal EEGNet [3] established a compact, generalisable CNN architecture for EEG-based BCIs using depthwise separable convolutions, achieving competitive performance across multiple BCI paradigms. Subsequent work introduced more specialised architectures, including ShallowConvNet and DeepConvNet which exploit temporal and spatial convolutions tailored to EEG signal structure. Despite their effectiveness, these architectures require continuous floating-point operations at every inference step, resulting in energy consumption of the order of 200 mJ per inference [4].

LSTM-based approaches [21] capture temporal dependencies in EEG sequences but at even greater computational cost, with training times on the PhysioNet dataset exceeding 15,000 seconds in comparative evaluations [21]. Hybrid CNN-LSTM models combine spatial feature extraction with temporal modelling but compound the energy limitations of both architectures.

3.2 Filter Bank CSP and Frequency Selection

The Filter Bank Common Spatial Patterns (FBCSP) method [7] established the standard for frequency band-based MI feature extraction, decomposing the EEG signal into multiple sub-bands and applying CSP spatial filtering within each band. Despite its effectiveness, FBCSP employs uniform, fixed sub-bands (4 Hz wide, 2 Hz step) identical across all subjects, failing to account for inter-individual spectral variability.

Recent work has begun to address this limitation. Sparrow Search Algorithm-based FBCSP (SSA-FBCSP) [19] achieved a 21.76% accuracy improvement over conventional fixed-band FBCSP by optimising the sub-band configuration per subject on BCI Competition IV-2a. Auto-selected regularised FBCSP [20] demonstrated improvements of 7.14% on BCI-2a and 9.54% on BCI-3a through data-driven band selection. These results provide strong empirical evidence for the superiority of adaptive, per-subject frequency selection over fixed-band approaches.

3.3 Spiking Neural Networks for EEG-BCI

SGLNet [8] represents the current state-of-the-art in SNN-based MI-EEG classification, integrating a learnable spike encoder with spike-based multi-head adaptive graph convolution (MHAGC) and spike-based LSTM. On the PhysioNet dataset, SGLNet achieved $84.15 \pm 0.58\%$ subject-mixed accuracy across 4-class MI tasks, demonstrating that SNN-based architectures can approach the performance of conventional deep learning BCIs.

HR-SNN [9] proposed an end-to-end SNN for 4-class MI classification, reporting 71.88% subject-specific accuracy on PhysioNet with 52–53% energy reduction compared to equivalent CNN implementations. RD-SNN [10] applied Roman domination-based optimisation to SNN training, achieving 73.65% accuracy on PhysioNet. SCNet [16] introduced adaptive learnable coding for MI-EEG classification and outperformed all fixed-encoding SNN approaches. AFTSC-Net [17] demonstrated that adaptive firing thresholds in SNNs simultaneously reduce computation and improve accuracy.

Despite these advances, all existing SNN-BCI systems share a fundamental architectural limitation: they employ a single membrane time constant, applying the same temporal dynamics to μ-band and β-band features that are neurophysiologically generated at different timescales.

3.4 Parametric LIF and Multi-Timescale SNNs

PLIF [11] introduced the Parametric Leaky Integrate-and-Fire neuron, in which the membrane time constant τ is made learnable per layer:

$$k = \sigma(a) = \frac{1}{1+e^{-a}}, \quad \tau = \frac{1}{k}, \quad v_j^{t+1} = (1-k)v_j^t + kI_j^t \tag{4}$$

where $a \in \mathbb{R}$ is a learnable scalar and $\sigma(\cdot)$ is the sigmoid function ensuring $\tau > 0$. PLIF consistently outperformed fixed-τ LIF on DVS Gesture, N-CALTECH, and N-MNIST benchmarks, demonstrating that temporal adaptability is critical for spike-based processing of temporal signals.

The e-prop framework [14] provided the theoretical foundation for multi-timescale recurrent SNNs, proving that networks with heterogeneous temporal constants — implemented through adaptive LIF (ALIF) neurons with slowly varying firing thresholds — can achieve LSTM-level computational performance on temporal tasks while maintaining biological plausibility. Li *et al.* [15] recently validated parallel multi-timescale SNN processing for BCI signals, demonstrating simultaneous improvements in accuracy and energy efficiency through multi-dimensional attention-weighted feature fusion.

3.5 EEGSNet and Adaptive Thresholding

EEGSNet [18] integrated PLIF neurons with a sample-based adaptive thresholding strategy, achieving a 5.49% improvement on cross-subject MI classification tasks compared to fixed-threshold baselines. This result provides direct empirical support for the learnable threshold approach proposed in this work. However, EEGSNet does not jointly optimise the encoding threshold with a sparsity penalty in a unified loss function, which constitutes a key differentiator of our proposed framework.

4. Research Gaps

Based on the preceding survey, we identify five critical and underaddressed research gaps in the current EEG-BCI literature.

4.1 Gap 1: Energy Inefficiency of Deep Learning BCIs

Conventional CNN and LSTM-based BCIs consume approximately 200 mJ per inference [4], which is approximately 40× the total power budget of wearable BCI devices (<5 mJ). While SNN-based BCIs offer a theoretical energy advantage through sparse spike-based computation, the actual energy savings achievable on neuromorphic hardware have been validated at up to 95% compared to DNN equivalents on EEG motor imagery tasks [5]. None of the existing SNN-BCI systems fully exploits this potential through explicit energy optimisation during training.

4.2 Gap 2: Subject-Agnostic Fixed Frequency Band Selection

All existing EEG-BCI systems — including the state-of-the-art SGLNet [8] — apply identical frequency bands to all subjects, ignoring well-documented inter-individual variability of 2–5 Hz in EEG spectral peaks [6]. Adaptive per-subject band selection methods have demonstrated improvements of up to 21.76% over fixed-band approaches [19], yet no existing SNN-BCI system incorporates subject-specific frequency adaptation.

4.3 Gap 3: Manually Engineered Spike Encoding Thresholds

The conversion of continuous EEG signals into discrete spike trains requires a threshold parameter θ. In existing SNN-BCIs, this threshold is either manually set (typically at the signal mean) or learned independently from the classification objective. Bauer *et al.* [5] identified inefficient spike encoding as the primary bottleneck in current SNN-BCI systems. No published work has jointly optimised the encoding threshold with a sparsity regularisation term in a unified end-to-end loss function on the PhysioNet 4-class MI task.

4.4 Gap 4: Single Temporal Scale in SNN-BCIs

Motor imagery EEG exhibits distinct dynamics at multiple timescales: fast μ-band transients (∼5 ms) at movement onset and slow β-band oscillations (∼30 ms) during sustained motor states [22]. All existing SNN-BCI systems employ a single membrane time constant, inherently limiting their ability to process these temporally heterogeneous signal components simultaneously. While PLIF [11] and e-prop [14] have established the theoretical and empirical foundations for multi-timescale SNN processing, no prior work has applied dual-branch PLIF architecture to non-invasive EEG motor imagery classification.

4.5 Gap 5: Absence of Standardised Energy Evaluation

A systematic survey of deep learning MI-EEG classification papers [21] found that the majority report only classification accuracy, with no energy measurement or computational cost analysis. Without standardised metrics, it is impossible to make fair comparisons of efficiency claims across systems. The Synaptic Operations (SynOps) metric [24,25] constitutes the mathematically validated standard for SNN energy proxy estimation, yet it is employed by very few published SNN-BCI systems.

5. Proposed Methodology

We propose an *Adaptive EEG-to-Spike Framework* comprising a six-stage pipeline that addresses all five identified research gaps through three novel contributions.

5.1 Stage 1: Signal Conditioning

Raw EEG recordings are pre-processed to remove artefacts and normalise the signal:

1. **Bad subject removal:** Six subjects with documented recording anomalies are excluded (S038, S088, S089, S092, S100, S104) [31], retaining 103 usable subjects.
2. **Bandpass filtering:** A 4th-order Butterworth filter with passband 0.5–40 Hz preserves physiologically relevant EEG activity while removing DC drift and high-frequency EMG noise.
3. **ICA artefact removal:** Independent Component Analysis with $n = 20$ components removes eye blink and muscle artefacts via automated EOG/EMG correlation detection.
4. **Bad trial rejection:** Epochs with amplitude exceeding $\pm 100\,\mu$V in the 8–30 Hz band are excluded.
5. **Baseline correction:** A -200 ms to 0 ms pre-stimulus baseline is subtracted to remove slow signal drift.
6. **Per-subject Z-score normalisation:** $x \leftarrow (x - \mu)/\sigma$, where μ and σ are computed *exclusively* on the training fold to prevent data leakage.

5.2 Stage 2: Spatial Filtering

Regularised Common Spatial Patterns (RCSP) with Ledoit-Wolf shrinkage [28] are applied to enhance class-discriminative spatial information:

$$\mathbf{W}^* = \arg\max_{\mathbf{W}} \frac{\mathbf{W}^\top \hat{\mathbf{\Sigma}}_1 \mathbf{W}}{\mathbf{W}^\top \hat{\mathbf{\Sigma}}_2 \mathbf{W}} \tag{5}$$

where the regularised covariance estimate is:

$$\hat{\mathbf{\Sigma}} = (1 - \alpha)\hat{\mathbf{\Sigma}}_{sample} + \alpha\bar{\sigma}\mathbf{I} \tag{6}$$

Standard CSP is extended to the 4-class setting via One-vs-Rest (OvR) decomposition. All spatial filter parameters are estimated exclusively on the training fold of each cross-validation split.

5.3 Stage 3: Adaptive Band Selection (Contribution 1)

A filter bank of 7 sub-bands with 4 Hz bandwidth and 2 Hz step covering the range 4–32 Hz is constructed. For each sub-band b, log-variance features are computed:

$$f_b = \log(\mathrm{Var}(x_b)) \tag{7}$$

Sub-band ranking is performed per subject using the KNN Mutual Information estimator of Kraskov *et al.* [27]:

$$\hat{I}(X;Y) = \psi(k) - \langle\psi(n_x+1)\rangle - \langle\psi(n_y+1)\rangle + \psi(N) \tag{8}$$

where $\psi(\cdot)$ is the digamma function, k is the number of nearest neighbours, and N is the number of samples. The KNN estimator is selected over histogram-based alternatives due to its consistency and low bias at small sample sizes ($N = 84$ trials per subject).

The top-$K = 3$ sub-bands per subject are selected and frozen prior to the training phase, ensuring the band selection stage operates exclusively on training data and does not introduce leakage.

5.4 Stage 4: Trainable Spike Encoding (Contribution 2)

The selected EEG features are encoded into spike trains using a threshold comparison applied over $T = 16$ timesteps (corresponding to 250 ms windows over a 4 s epoch):

$$S(t) = H(x(t) - \theta), \quad H(u) = \begin{cases} 1 & u > 0 \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

The threshold θ is initialised at the 75th percentile of the absolute signal amplitude on the training set, providing approximately 25% spike density at initialisation. A hard lower bound $\theta \geq \epsilon > 0$ prevents threshold collapse.

Unlike all prior SNN-BCI work, θ is treated as a learnable parameter and optimised jointly within the unified training objective:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda(e) \sum_{t=1}^{T} S(t) \tag{10}$$

where $\mathcal{L}_{CE}$ is the cross-entropy classification loss and $\lambda(e)$ implements a linear warm-up schedule: $\lambda = 0$ for the first 20 epochs, increasing linearly to λ_{max} over epochs 21–40. This joint optimisation directly links the training objective to the SynOps energy proxy (Eq. 2), as minimising $\sum S(t)$ is equivalent to minimising the total SynOps of the network.

5.5 Stage 5: Multi-Timescale PLIF Network (Contribution 3)

The encoded spike trains are processed by a dual-branch PLIF network. The membrane dynamics of each PLIF neuron follow Eq. (4), with the learnable parameter a initialised differently in each branch:

- **Fast branch:** a_{init} set such that $\tau \approx 5$ ms, specialised for rapid μ-band transients.
- **Slow branch:** a_{init} set such that $\tau \approx 30$ ms, specialised for sustained β-band oscillations.

Both branches process the identical spike input tensor in parallel. Their outputs are fused via an attention mechanism:

$$h = \alpha h_{fast} + (1 - \alpha) h_{slow}, \quad \alpha = \sigma(w) \tag{11}$$

where $w \in \mathbb{R}$ is a learnable scalar. An entropy regularisation term is added to the loss to prevent attention collapse to a degenerate single-branch solution:

$$\mathcal{L}_{ent} = -\alpha \log \alpha - (1 - \alpha) \log(1 - \alpha) \tag{12}$$

Batch normalisation is applied after the spike encoder output and Dropout ($p = 0.5$) is applied after the attention fusion layer.

5.6 Stage 6: Classification and Evaluation

The fused representation is passed to a classification head that performs spike voting over the $T = 16$ timesteps. The final loss function combines the classification objective, sparsity penalty, and entropy regularisation:

$$\mathcal{L}_{total} = \mathcal{L}_{MSE} + 0.1 \cdot \mathcal{L}_{CE} + \lambda(e) \sum_{t=1}^{T} S(t) + \beta \mathcal{L}_{ent} \tag{13}$$

where $\mathcal{L}_{MSE}$ is the mean squared error between output spike counts and one-hot class targets, and β is a small regularisation coefficient for the entropy term.

6. Evaluation Protocol

6.1 Dataset

Experiments are conducted on the PhysioNet EEG Motor Movement/Imagery Dataset [30], comprising 109 subjects performing imagined and executed movements of the left hand, right hand, both feet, and rest. Following standard preprocessing practice [31], six subjects with recording anomalies are excluded, leaving 103 subjects. Each subject provides 64-channel EEG recorded at 160 Hz.

6.2 Validation Protocol

1. **Subject-mixed 5-fold CV:** Trials from all subjects are pooled and divided into 5 folds. All preprocessing and band selection operations are performed inside the CV loop to prevent data leakage.
2. **Leave-One-Subject-Out (LOSO):** Evaluated on a randomly selected subset of 10 subjects to assess cross-subject generalisation.

6.3 Evaluation Metrics

The following metrics are reported:

- **Classification accuracy** (%)
- **Cohen's Kappa** (κ): corrects for chance-level agreement
- **F1-score per class**: evaluates class-level discriminability
- **SynOps count**: computed via Eq. (2), reported as a percentage reduction relative to the EEGNet CNN baseline

6.4 Ablation Study

A four-condition ablation study isolates the contribution of each proposed component:

1. **Baseline:** Fixed bands, fixed θ, standard LIF
2. **+Bands:** Adaptive KNN-MI bands, fixed θ, standard LIF
3. **+Encoding:** Fixed bands, learnable θ with sparsity loss, standard LIF
4. **Full model:** All three contributions combined

6.5 Comparison Baselines

The proposed framework is compared against four published baselines:

- **EEGNet** [3]: compact CNN; universal BCI baseline
- **SGLNet** [8]: SNN with graph convolution and LSTM; 84.15% on PhysioNet
- **HR-SNN** [9]: end-to-end SNN; 71.88% on PhysioNet
- **RDSNN** [10]: optimised SNN; 73.65% on PhysioNet

7. Discussion

7.1 Expected Contributions

The proposed framework makes three specific measurable contributions over the current state-of-the-art:

Contribution 1 (Adaptive bands): Per-subject KNN-MI band selection is expected to yield accuracy improvements consistent with the +21.76% reported by SSA-FBCSP [19] on BCI Competition IV-2a, where the same underlying principle of subject-specific spectral adaptation was validated.

Contribution 2 (Learnable encoding): Joint optimisation of θ with a sparsity penalty constitutes a novel contribution on the PhysioNet 4-class MI task. EEGSNet [18] demonstrated +5.49% cross-subject improvement through adaptive thresholding without the joint sparsity term, supporting the hypothesis that learnable encoding yields measurable performance gains.

Contribution 3 (Multi-timescale PLIF): The dual-branch fast/slow architecture is grounded in the PLIF benchmark [11], which demonstrated consistent superiority of learnable-τ neurons over fixed-τ counterparts across all evaluated temporal classification tasks.

7.2 Energy Efficiency Expectations

SynOps reduction of $> 30\%$ over the EEGNet baseline is expected based on the inherent sparsity of SNN computation. The spike-count regularisation term in Eq. (10) provides an additional training-time mechanism to further reduce spike density, consistent with the 84% SynOps reduction reported by Perez-Nieves and Bhatt [26] using analogous regularisation strategies.

7.3 Limitations

Several limitations of the proposed framework are acknowledged:

- **Software energy proxy:** SynOps counting provides a reproducible energy proxy but does not constitute a direct hardware power measurement. Validation on neuromorphic hardware (Intel Loihi or SpiNNaker) is required for real-world deployment claims and is deferred to future work.
- **Single dataset:** Evaluation is restricted to the PhysioNet dataset. Validation on BCI Competition IV-2a would strengthen generalisability claims.
- **Offline processing:** The framework operates on pre-recorded EEG epochs and does not address real-time online BCI deployment constraints.
- **Surrogate gradient choice:** The arctangent surrogate is adopted following SpikingJelly defaults; systematic comparison with alternative surrogates (exponential, piecewise-linear) is not performed.

8. Conclusion

This paper has presented a systematic survey of EEG-based motor imagery BCI systems, identifying five critical research gaps that collectively prevent existing SNN-based approaches from achieving the dual objectives of high classification accuracy and genuine energy efficiency. To address these gaps, we proposed the Adaptive EEG-to-Spike Framework incorporating three novel contributions: per-subject KNN-MI adaptive frequency band selection, end-to-end learnable spike encoding with joint sparsity regularisation, and a dual-branch multi-timescale PLIF network with attention fusion.

Each contribution is supported by quantitative benchmark evidence from peer-reviewed literature

(2020–2025), with adaptive band selection demonstrating up to 21.76% accuracy improvement, learnable thresholding achieving +5.49% cross-subject improvement, and SNN deployments on neuromorphic hardware achieving 95% energy reduction. The proposed SynOps-based evaluation protocol provides a standardised, reproducible mechanism for comparing energy efficiency across heterogeneous SNN-BCI systems.

Future work will focus on hardware deployment on Intel Loihi for direct energy measurement, multi-dataset validation, and online real-time BCI adaptation using the proposed framework.

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